

Sumedh Kothari

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EDUCATION

Georgia Institute of Technology

Expected May 2028

B.S. in College of Engineering

- **Coursework:** Computer Organization and Programming; Computing for Engineers; Advanced Data Structures, Embedded Systems, and Networking; Ordinary Differential Equations
- **Honors:** Symposium of Rising Scholars Presenter, Eagle Scout

EXPERIENCE

Georgia Tech Renewable Bioproducts Institute | Tong Laboratory

Atlanta, GA

Research Intern

August 2025 – Present

- Used Python, NumPy/Pandas libraries to apply Bayesian statistical modeling to optimize lignin valorization reaction
- Utilized NMR, FTIR, and DFT modeling data to characterize value-added chemicals from lignin breakdown
- Completed comprehensive literature reviews on aromatic compound synthesis, 3D printing custom GO structures, and optimal catalyst design while collaborating with research team to write manuscript for publication

Hydrogen Engineering Research Consortium

Los Angeles, CA

Simulation Intern

August 2024 – February 2025

- Used MATLAB, dimensionless differential equations to model species' concentrations in eutrophic freshwater ecosystem
- Employed sustainability over sets calculation to determine equilibria, stability of environment; co-authored research paper on environmental systems modeling under UCLA faculty mentorship

PROJECTS

IEEE Robotech 2026 Hackathon - Autonomous Track | Georgia Tech

Atlanta, GA

Mechanical/Software Engineer

January 2026

- Designed and CAD-modeled a fully autonomous debris-clearing robot by engineering the chassis, actuation mechanisms, and beacon-deployment system, enabling end-to-end autonomous operation without human intervention
- Trained, developed, and deployed a YOLO-based computer vision model in PyTorch by integrating it into the onboard perception pipeline, achieving 95% debris detection accuracy and enabling real-time hazard identifications
- Integrated perception, navigation, and mechanical actuation systems by linking CV outputs to motion planning and debris-removal hardware, allowing the robot to autonomously clear obstacles to certify landing safety

Sports Betting Arbitrage Calculator | Python, REST APIs

Atlanta, GA

Developer

November 2025

- Developed application that aggregates live odds from multiple sportsbooks via API, normalizes data, and identifies risk-free arbitrage opportunities across multiple market types (H2H, Spread, Total) with up to 50% daily ROI.
- Built a robust, production-ready pipeline with Dockerization, session management, error handling, and logging, enabling scalable, real-time analysis across dozens of active sports markets.

AiApply - AI Job Applier | Python, Selenium WebDriver, HTML, JavaScript

Atlanta, GA

Developer

October 2025

- Developed full-stack AI-powered internship application platform to automate end-to-end job search workflows, including resume optimization, cover letter generation, and programmatic application submission on Handshake
- Implemented Selenium-based automation to extract recruiter profiles from filtered company searches and deliver personalized direct messages at scale, increasing outreach efficiency and response rates
- Engineered web scraping algorithms to identify target companies based on location and industry parameters, then automated tailored email outreach highlighting candidate skillsets parsed from resume data

GreenGrade - Gemini API Developer Competition | Flutter, C++

Los Angeles, CA

Project Lead

July 2024 - August 2025

- Directed 20-person UCLA team to architect and implement a fuzzy logic-driven Sustainability Interval Index calculator with scalable data integration pipelines
- Developed and deployed *GreenGrade*, embedding SII computational model and automating sustainability analytics for 175 mining firms through Gemini API

SKILLS

- **Technical:** Python, Java, Data analysis (Excel, MATLAB), CAD (SolidWorks)
- **Communication:** Technical Writing, Multilingual (English, Hindi, Spanish)